

Green Spaces Improvement Project

Project Overview

The Green Spaces Improvement Project is a key initiative for Willow Creek. It is designed to strengthen environmental quality and improve everyday quality of life for residents. Launched on November 10, 2023 with a budget of \$450,000, the project supports the town's broader climate-adaptation strategy and its commitment to well-being in the urban environment.

Context and Rationale

Willow Creek currently manages about 37 acres (15 hectares) of green spaces across parks, neighborhood squares, and public gardens. A detailed review identified several issues that need coordinated action:

- aging trees and landscape features;
- outdated layouts that no longer meet current climate-resilience needs;
- limited biodiversity and fragmented ecological corridors;
- high water use for maintenance;
- insufficient accessibility for all users;
- a shortage of play, recreation, and fitness amenities;
- no consistent landscape identity across town.

Given these findings and current environmental challenges, the town must rethink both the design and management of its green spaces so they become true multifunctional green lungs for the community.

Project Objectives

The redevelopment of municipal parks and gardens has several complementary objectives:

- Strengthen climate resilience by creating urban cooling areas.
- Support biodiversity through more diverse habitats and plant species.
- Optimize resource management, especially water and energy.
- Improve accessibility for people of all ages and mobility levels.
- Develop recreational, athletic, cultural, and educational uses.
- Create social connection through shared activity and gathering spaces.
- Highlight the town's landscape identity by building on local natural heritage.
- Involve residents in the design and ongoing care of green spaces.

Sites Covered and Scope of Work

The project covers five main sites across Willow Creek:

1. Central Park — 12.4 acres (5 hectares)

- full redesign of paths and plantings;
- creation of a wetland area and a botanical garden;
- installation of play areas and urban sports equipment;
- development of a multipurpose kiosk.

2. Jean Moulin Square — 2 acres (0.8 hectare)

- conversion into a rain garden with landscaped bioswales;
- installation of ecological water-play features;
- creation of a heritage orchard.

3. Ramparts Gardens — 3 acres (1.2 hectares)

- rehabilitation of dry-stone walls;
- installation of themed gardens for aromatic and medicinal plants;
- creation of a small outdoor performance lawn.

4. Willow Greenway — 6.2 acres (2.5 hectares)

- reinforcement of creek banks;
- creation of a fitness and educational trail;
- development of protected biodiversity areas.

5. School Square — 1.2 acres (0.5 hectare)

- conversion into an educational garden;
- installation of shared vegetable plots;
- creation of a wildlife observation area.

Technical Description of Improvements

Planting and Greening

- planting 250 trees from diverse species adapted to climate change;
- creating 37,675 sq ft (3,500 m²) of pollinator-friendly flowering meadows;
- installing 12,915 sq ft (1,200 m²) of low-water shrub beds;
- planting 4,500 local drought-resistant perennials;
- creating approximately 2,625 linear feet (800 meters) of mixed native hedgerows;
- creating two Miyawaki-style micro-forests.

Water Management

- installing five landscaped retention basins with a combined capacity of about 15,900 cu ft (450 m³);
- creating permeable surfaces and bioswales to reduce runoff;
- using drought-tolerant planting plans to reduce irrigation needs;
- promoting rainwater capture and reuse wherever technically feasible.

Expected Benefits

The project is expected to reduce heat-island effects, improve stormwater management, increase biodiversity, and make public spaces more welcoming for residents. It also creates new educational and recreational uses that support community life.

Conclusion

By redesigning its parks, gardens, and green corridors, Willow Creek is investing in a healthier, more resilient, and more inclusive public realm. The project will help residents reconnect with nature while preparing the town for long-term climate challenges.